

PS-11 — your data model

WanderMatch — Social & Group Travel Planning

Kognivera Hackathon 2026 · Travel & Tourism · data model v1.1.0-rc1

The problem statement itself, the 24-hour MVP scope and the XR device requirement live in the hackathon application, on your statement's page. This document is the data you have been given to build it with: every table, every field, and what each one is for.

You have **27,428 rows across 14 tables**. 13 of them are the tables this statement is built on; the remaining 1 is a reference table the others point at, included so the database works on its own.

All of it is in the `data/` folder beside this document: as `PS-11.db` (SQLite, indexed, ready to query), as CSV, and as DDL for Postgres and SQLite.

What the data gives you

270 group trips with real membership and roles, 274 live proposals, 761 votes — including genuine ties — and versioned itineraries so conflict handling is tractable.

Watch out for this one

`itineraries.version` is how you handle two people editing at once. Bump it, do not overwrite it.

The tables this statement is built on

TABLE	ROWS	WHAT YOU USE IT FOR
<code>cities</code>	60	The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.
<code>countries</code>	30	ISO country reference. Every city, currency default and calling code resolves here.
<code>currencies</code>	25	carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).
<code>itineraries</code>	803	A versioned plan belonging to a trip. version is what makes PS-11's conflict handling tractable.
<code>itinerary_items</code>	8,583	The atom of the portal, and the single most-shared object across the thirteen builds. If a team implements one shared shape, this is it.
<code>trip_members</code>	1,407	Membership and role. PS-11's collaborative editing and PS-08's split lines both key off this.

TABLE	ROWS	WHAT YOU USE IT FOR
trips	600	The container that gives dates, party, destination and budget to everything else. Seven statements produce or consume one.
user_preferences	1,200	Explicit preference signal. PS-04's language-preference requirement reads from here.
users	1,200	The traveller identity every personalisation hangs off. Segmented heavy / light / cold_start so APS-04 can prove cold start.
proposals	274	Pre-existing group debate, so PS-11 has something to demo the moment it loads.
tour_guides	120	PS-04's added guide dimension — selectable by language, specialisation, availability and price.
user_interactions	12,339	heavy users with long histories plus a deliberate cold-start cohort with none. One event per user makes everyone cold and APS-04 undemonstrable.
votes	761	One row per member per proposal. Ties are deliberately present in the seed so consensus logic gets exercised.

Reference tables, included so the database is valid

You will mostly join through these rather than think about them.

TABLE	ROWS	WHAT IT IS
languages	26	Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.

How they fit together

Open `02_DATA_MODEL_DIAGRAM.html` in a browser for the clickable version — it shows these tables and nothing else. Download it first; it will not render inside SharePoint.

Some tables point at "any bookable thing" using an `(entity_type, entity_id)` pair rather than a typed foreign key. That is deliberate: it is what lets one feature refer to a hotel, a flight, a point of interest or a package without a separate join table for each. The legal values of `entity_type` are in `data/enums.json`.

Every field, table by table

Columns marked **PK** are the primary key. **FK** shows what a column points at. Enum columns list their legal values — anything else is rejected by the conformance check.

currencies

Reference & geography · 25 rows · IDs start `cur_`

carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).

COLUMN	TYPE	CONSTRAINTS	NOTES
currency_id	text	PK	cur_ prefixed.
iso4217	char(3)	UNIQUE · NOT NULL	e.g. INR.
name	text	NOT NULL	
symbol	text	NOT NULL	
minor_unit_exponent	smallint	NOT NULL	0 for JPY/KRW, 2 default, 3 for KWD/BHD.
display_locale	text	NOT NULL	BCP-47 locale used for formatting.
updated_at	timestamptz	NOT NULL	

languages

Reference & geography · 26 rows · IDs start lng_ · reference table

Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.

COLUMN	TYPE	CONSTRAINTS	NOTES
language_id	text	PK	lng_ prefixed.
bcp47	text	UNIQUE · NOT NULL	e.g. ta, hi, en-IN — never 'Tamil'.
english_name	text	NOT NULL	
native_name	text	NOT NULL	
script	text	NOT NULL	ISO-15924, e.g. Taml, Deva, Latn.
rtl	bool	NOT NULL	Right-to-left rendering flag.
tts_supported	bool	NOT NULL	Relevant to PS-13 voice output.
updated_at	timestamptz	NOT NULL	

countries

Reference & geography · 30 rows · IDs start cnt_

ISO country reference. Every city, currency default and calling code resolves here.

COLUMN	TYPE	CONSTRAINTS	NOTES
country_id	text	PK	Canonical ID, cnt_ prefixed.
iso2	char(2)	UNIQUE · NOT NULL	ISO-3166-1 alpha-2, e.g. IN.
iso3	char(3)	UNIQUE · NOT NULL	ISO-3166-1 alpha-3, e.g. IND.
name	text	NOT NULL	English short name.

COLUMN	TYPE	CONSTRAINTS	NOTES
default_currency	char(3)	FK → currencies.iso4217 · NOT NULL	ISO-4217 code.
calling_code	text	NOT NULL	E.164 country calling code, e.g. +91.
region	text	NOT NULL	UN sub-region grouping.
updated_at	timestampz	NOT NULL	Rule R4: UTC, ISO-8601 with offset.

cities

Reference & geography · 60 rows · IDs start `cty_`

The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.

COLUMN	TYPE	CONSTRAINTS	NOTES
city_id	text	PK	cty_ prefixed.
name	text	NOT NULL	City name.
state	text		State / province, nullable for city-states.
country_id	text	FK → countries.country_id · NOT NULL	
country_code	char(2)	NOT NULL	Denormalised ISO2 for convenient joins.
lat	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
lng	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
timezone	text	NOT NULL	IANA zone, e.g. Asia/Kolkata.
region	text	NOT NULL	Domestic region grouping, e.g. South India.
population	int		Approximate, for demand weighting.
season_profile	text	NOT NULL · one of winter, summer, monsoon, post_monsoon, spring, autumn	Dominant season at the peak travel window.
peak_months	text	NOT NULL	Comma-separated month numbers, e.g. 10,11,12.
primary_language	text	FK → languages.bcp47 · NOT NULL	Rule R6: BCP-47 tag.
description	text		One-paragraph orientation blurb, used by

COLUMN	TYPE	CONSTRAINTS	NOTES
			PS-13.
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

tour_guides

Supply & catalogue · 120 rows · IDs start gid_

PS-04's added guide dimension — selectable by language, specialisation, availability and price.

COLUMN	TYPE	CONSTRAINTS	NOTES
guide_id	text	PK	gid_ prefixed.
city_id	text	FK → cities.city_id · NOT NULL	
display_name	text	NOT NULL	Synthetic.
languages	text	NOT NULL	Comma-separated BCP-47 — the PS-04 filter.
specialisation	text	NOT NULL · one of heritage, food, trekking, wildlife, photography, religious, shopping, accessibility	
secondary_specialisation	text	one of heritage, food, trekking, wildlife, photography, religious, shopping, accessibility	
years_experience	smallint	NOT NULL	
rating	decimal(2,1)		Null for new guides.
review_count	int	NOT NULL	
day_rate	decimal(12,2)	NOT NULL	D3.
half_day_rate	decimal(12,2)	NOT NULL	
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
certified	bool	NOT NULL	
bio	text	NOT NULL	Plausible prose.
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

users

Identity & preference · 1,200 rows · IDs start `usr_`

The traveller identity every personalisation hangs off. Segmented heavy / light / cold_start so APS-04 can prove cold start.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>user_id</code>	text	PK	usr_ prefixed.
<code>display_name</code>	text	NOT NULL	Synthetic — no real people (content policy).
<code>email</code>	text	UNIQUE · NOT NULL	Synthetic @example.invalid addresses only.
<code>home_city_id</code>	text	FK → <code>cities.city_id</code> · NOT NULL	
<code>home_currency</code>	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	
<code>locale</code>	text	FK → <code>languages.bcp47</code> · NOT NULL	UI language, BCP-47.
<code>budget_band</code>	text	NOT NULL · one of <code>shoestring</code> , <code>value</code> , <code>mid</code> , <code>premium</code> , <code>luxury</code>	
<code>travel_style</code>	text	NOT NULL · one of <code>budget</code> , <code>comfort</code> , <code>luxury</code> , <code>adventure</code> , <code>slow</code> , <code>cultural</code> , <code>wellness</code>	
<code>traveller_type</code>	text	NOT NULL · one of <code>solo</code> , <code>couple</code> , <code>family</code> , <code>business</code> , <code>friends</code> , <code>senior</code> , <code>backpacker</code>	
<code>segment</code>	text	NOT NULL · one of <code>heavy</code> , <code>light</code> , <code>cold_start</code>	heavy / light / cold_start cohorts.
<code>date_of_signup</code>	date	NOT NULL	
<code>loyalty_tier</code>	text		none
<code>status</code>	text	NOT NULL · one of <code>active</code> , <code>inactive</code> , <code>archived</code> , <code>draft</code>	
<code>created_at</code>	timestampz	NOT NULL	
<code>updated_at</code>	timestampz	NOT NULL	

trips

Trip & itinerary · 600 rows · IDs start `trp_`

The container that gives dates, party, destination and budget to everything else. Seven statements produce or consume one.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>trip_id</code>	text	PK	trp_ prefixed.
<code>owner_user_id</code>	text	FK → <code>users.user_id</code> · NOT NULL	

COLUMN	TYPE	CONSTRAINTS	NOTES
title	text	NOT NULL	
origin_city_id	text	FK → cities.city_id	
destination_city_id	text	FK → cities.city_id · NOT NULL	
start_date	date	NOT NULL	Rule R4: zoneless calendar date.
end_date	date	NOT NULL	
party_size	smallint	NOT NULL	
adults	smallint	NOT NULL	
children	smallint	NOT NULL	
trip_type	text	NOT NULL · one of solo, couple, family, business, friends, senior, backpacker	
is_group_trip	bool	NOT NULL	PS-11 / PS-08 filter.
status	text	NOT NULL · one of draft, planning, confirmed, in_progress, completed, cancelled	
home_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
notes	text		
created_at	timestamptz	NOT NULL	
updated_at	timestamptz	NOT NULL	

user_interactions

Signals & evaluation · 12,339 rows · IDs start uix_

heavy users with long histories plus a deliberate cold-start cohort with none. One event per user makes everyone cold and APS-04 undemonstrable.

COLUMN	TYPE	CONSTRAINTS	NOTES
interaction_id	text	PK	uix_ prefixed.
user_id	text	FK → users.user_id · NOT NULL	
entity_type	text	NOT NULL · one of hotel, room_type, rate_plan, flight, flight_fare, poi, package, package_component, guide, transfer, event, xr_scene	
entity_id	text	NOT NULL	

COLUMN	TYPE	CONSTRAINTS	NOTES
interaction_type	text	NOT NULL · one of view, click, like, save, book, dismiss, share, search	
occurred_at	timestampz	NOT NULL	
dwell_seconds	int		Null for non-view events.
position_in_list	smallint		Rank at which the item was shown — needed for unbiased offline eval.
query_text	text		Set for interaction_type = search.
query_language	text	FK → languages.bcp47	
channel	text	NOT NULL · one of web, mobile_app, partner, call_centre, agent	
session_id	text	NOT NULL	
implicit_rating	decimal(3,2)		Derived signal, provided for convenience.

user_preferences

Identity & preference · 1,200 rows · IDs start `prf_`

Explicit preference signal. PS-04's language-preference requirement reads from here.

COLUMN	TYPE	CONSTRAINTS	NOTES
preference_id	text	PK	prf_ prefixed.
user_id	text	FK → users.user_id · UNIQUE · NOT NULL	One row per user.
preferred_languages	text	NOT NULL	Comma-separated BCP-47 tags, most-preferred first.
guide_language	text	FK → languages.bcp47	PS-04 — preferred language for guide/tour delivery.
interests	text	NOT NULL	Comma-separated category codes.
dietary_flags	text		vegetarian
accessibility_needs	text		step_free
preferred_currency	char(3)	FK → currencies.iso4217 · NOT NULL	
max_daily_budget	decimal(12,2)		decimal, paired with currency below.

COLUMN	TYPE	CONSTRAINTS	NOTES
max_daily_budget_currency	char(3)	FK → currencies.iso4217	
pace	text	NOT NULL	relaxed
updated_at	timestamptz	NOT NULL	

itineraries

Trip & itinerary · 803 rows · IDs start `itn_`

A versioned plan belonging to a trip. version is what makes PS-11's conflict handling tractable.

COLUMN	TYPE	CONSTRAINTS	NOTES
itinerary_id	text	PK	itn_ prefixed.
trip_id	text	FK → trips.trip_id · NOT NULL	
name	text	NOT NULL	
version	int	NOT NULL	Monotonic per itinerary.
is_active	bool	NOT NULL	Exactly one active version per trip.
generated_by	text	NOT NULL · one of user , ai_planner , optimizer , agent , vote , import	Which subsystem produced this version.
total_cost	decimal(12,2)	NOT NULL	sum of item costs, half-up at the end.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
total_duration_minutes	int	NOT NULL	
total_carbon_kg	decimal(10,3)	NOT NULL	APS-09 objective.
optimizer_weights	text		JSON string: {cost, time, carbon} weights that produced it.
status	text	NOT NULL · one of active , inactive , archived , draft	
created_at	timestamptz	NOT NULL	
updated_at	timestamptz	NOT NULL	

itinerary_items

Trip & itinerary · 8,583 rows · IDs start `itm_`

The atom of the portal, and the single most-shared object across the thirteen builds. If a team implements one shared shape, this is it.

COLUMN	TYPE	CONSTRAINTS	NOTES
item_id	text	PK	itm_ prefixed.
itinerary_id	text	FK → itineraries.itinerary_id · NOT NULL	
day_index	smallint	NOT NULL	1-based.
sort_order	smallint	NOT NULL	Order within the day.
starts_at	timestamptz		Rule R4: offset-carrying; null for unscheduled items.
ends_at	timestamptz		
item_type	text	NOT NULL · one of hotel, flight, poi, package, guide, transfer, meal, free	
entity_type	text	one of hotel, room_type, rate_plan, flight, flight_fare, poi, package, package_component, guide, transfer, event, xr_scene	Polymorphic supply reference.
entity_id	text		Canonical ID of the referenced supply row.
title	text	NOT NULL	
cost	decimal(12,2)	NOT NULL	D3.
currency	char(3)	FK → currencies.iso4217 · NOT NULL	
carbon_kg	decimal(8,3)	NOT NULL	
duration_minutes	int	NOT NULL	
source	text	NOT NULL · one of user, ai_planner, optimizer, agent, vote, import	user
explanation	text		Where APS-04 and PS-01 surface reasoning without a parallel structure.
locked	bool	NOT NULL	APS-09 hard constraint: must-see items cannot be dropped.
status	text	NOT NULL · one of proposed, confirmed, removed, replaced	Rule R8: removed items stay visible.
created_at	timestamptz	NOT NULL	
updated_at	timestamptz	NOT NULL	

proposals

Trip & itinerary · 274 rows · IDs start `prp_`

Pre-existing group debate, so PS-11 has something to demo the moment it loads.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>proposal_id</code>	text	PK	prp_ prefixed.
<code>itinerary_id</code>	text	FK → <code>itineraries.itinerary_id</code> · NOT NULL	
<code>proposed_by_user_id</code>	text	FK → <code>users.user_id</code> · NOT NULL	
<code>action</code>	text	NOT NULL	add
<code>target_item_id</code>	text	FK → <code>itinerary_items.item_id</code>	Null for an add.
<code>entity_type</code>	text	one of <code>hotel</code> , <code>room_type</code> , <code>rate_plan</code> , <code>flight</code> , <code>flight_fare</code> , <code>poi</code> , <code>package</code> , <code>package_component</code> , <code>guide</code> , <code>transfer</code> , <code>event</code> , <code>xr_scene</code>	
<code>entity_id</code>	text		
<code>title</code>	text	NOT NULL	
<code>rationale</code>	text		
<code>cost_delta</code>	decimal(12,2)	NOT NULL	
<code>currency</code>	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	
<code>closes_at</code>	timestampz	NOT NULL	
<code>status</code>	text	NOT NULL · one of <code>open</code> , <code>accepted</code> , <code>rejected</code> , <code>expired</code>	
<code>created_at</code>	timestampz	NOT NULL	
<code>updated_at</code>	timestampz	NOT NULL	

trip_members

Trip & itinerary · 1,407 rows · IDs start `tmb_`

Membership and role. PS-11's collaborative editing and PS-08's split lines both key off this.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>member_id</code>	text	PK	tmb_ prefixed.
<code>trip_id</code>	text	FK → <code>trips.trip_id</code> · NOT NULL	
<code>user_id</code>	text	FK → <code>users.user_id</code> · NOT NULL	
<code>role</code>	text	NOT NULL · one of <code>owner</code> , <code>editor</code> , <code>viewer</code>	

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>joined_at</code>	timestampz	NOT NULL	
<code>share_weight</code>	decimal(6,3)	NOT NULL	Default 1.000; custom splits use this.
<code>invited_by_user_id</code>	text	FK → <code>users.user_id</code>	
<code>status</code>	text	NOT NULL · one of <code>active</code> , <code>inactive</code> , <code>archived</code> , <code>draft</code>	Rule R8: departed members stay visible.
<code>updated_at</code>	timestampz	NOT NULL	

Unique together: (`trip_id`, `user_id`)

`votes`

Trip & itinerary · 761 rows · IDs start `vot_`

One row per member per proposal. Ties are deliberately present in the seed so consensus logic gets exercised.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>vote_id</code>	text	PK	<code>vot_</code> prefixed.
<code>proposal_id</code>	text	FK → <code>proposals.proposal_id</code> · NOT NULL	
<code>user_id</code>	text	FK → <code>users.user_id</code> · NOT NULL	
<code>value</code>	text	NOT NULL · one of <code>yes</code> , <code>no</code> , <code>abstain</code>	
<code>weight</code>	decimal(4,2)	NOT NULL	Default 1.00; owner votes may be weighted.
<code>comment</code>	text		
<code>cast_at</code>	timestampz	NOT NULL	
<code>updated_at</code>	timestampz	NOT NULL	

Unique together: (`proposal_id`, `user_id`)

The rules that apply to these fields

#	RULE
R1	Additive only. Add columns, tables and stores freely. Never rename, drop or repurpose a field that came with the data.
R2	IDs are opaque prefixed strings — <code>htl_a91f3c</code> . Never integers, never parsed for meaning.
R3	Money is a pair: a 2-place decimal plus an ISO-4217 currency code. Never a float.

#	RULE
R4	Time is ISO-8601 with an offset. <code>_at</code> fields carry an offset; <code>_date</code> fields have no zone.
R5	Enums are lowercase snake_case and the legal values are in <code>data/enums.json</code> .
R6	Language is a BCP-47 tag — <code>ta</code> , not "Tamil".
R7	Geography is WGS-84 to 6 decimal places, <code>lat</code> and <code>lng</code> together or not at all.
R8	Nothing is hard-deleted. Rows carry <code>status</code> and <code>updated_at</code> .

Add whatever you like beside these fields — new columns, new tables, your own vector store, your own services. That is the point of R1. What you must not do is rename or re-key the fields that came with the data, because that is what would stop sixteen independent builds being put together afterwards.

`data/WORKING_WITH_THE_DATA.md` has the loading instructions, including how to read money without corrupting it. `tools/validate_conformance.py` tells you in thirty seconds whether you are still conformant.